HALEY PATEL

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EDUCATION

New Jersey Institute of Technology, *Honor's Scholar* — Rising Junior, GPA: **3.92/4.0**

B.S. Computer Science *Minor*: Applied Mathematics, *Ying Wu College of Computing* — Expected Graduation: **May 2025**

- Awarded full honor's merit scholarship for all years of attendance. Dedicated 100+ hours to local community service.
- **Relevant Coursework:** Data Structures and Algorithms, Database Systems, Java, C/C++, Python, Statistics, Multivariable Calculus, Actuarial Mathematics, Programming in Linux (in progress), Discrete Mathematics, Quantitative Analysis

EXPERIENCE

Machine Learning for Advanced Manufacturing, *NJIT Honor's Summer Research Institute* — ***May 2023 – Present***

- Transformed **9 million data points** to produce an iterative machine learning algorithm that simulates and predicts temperature profiles of particles in a laser manufacturing process with **99.97%** accuracy.
- Utilized an Anaconda environment for the seamless use of NumPy, Pandas, Scikit-Learn, and PyTorch Python libraries.

Software Development Intern, *UPS (United Parcel Service)* — ***June 2023 – Present***

- Spearheaded phase one of an employee user experience research project by gathering and analyzing application usage data to pinpoint inefficiencies in employee workflows and perform sentiment analysis using Python, PowerBI, and Excel.
- Assigned to develop a full-stack employee email server management website to increase the efficiency of migrating employee email servers across regions. Created a Jira and Confluence board to track development progress.

PROJECTS

Banking Website and Database* [MySQL, PHP, HTML5, CSS, Oracle SQL Server]** — ***February – May 2023

- Designed and created a database on Oracle SQL Server while adhering to industry-standard database normalization rules to meet the specific operational requirements of a fictional banking institution.
- Utilized PHP and MySQL to establish the backend infrastructure for the banking website and perform CRUD operations.
- Designed with HTML5 and CSS, the website includes login pages, customer transaction features, and an employee management dashboard.

Task Organization Application* [RESTful API, PostgreSQL, Node.js, Express, React]** — ***June 2023

- Developed full-stack, user-friendly task list application by developing a RESTful API using Node.js and Express to enable seamless communication between application layers and perform CRUD operations in the PostgreSQL database.
- Used React hooks to handle state management and asynchronous operations for rendering

Compiler for SPL Language* [C++]** — ***February – April 2023

- Developed a compiler for a Simple Perl-Like Language through building a lexical analyzer and parser interpreter in C++.
- The compiler parses and interprets code written in the SPL language, producing correct program outputs or associated error messages.

RSA Encryption Implementation* [Python]** — ***May 2023

- Implemented RSA encryption by programming modular arithmetic functions and Euler's Totient Function from scratch.
- Improved runtime and computational cost of encrypting large inputs by 200% by implementing a bitwise shifting method.

LEADERSHIP AND EXPERIENCES

Vice President, *Business Masterminds* — ***January 2022 - Present***

- Initiated an annual, interactive Artificial Intelligence in Business workshop of 60+ attendees to provide education for leveraging artificial intelligence tools to innovate business solutions and assist in entrepreneurial/start-up strategies.
- Maintained connections with industry professionals and school alumni to serve as guest speakers in workshops/events.

Financial Analyst, *Investment Fund NJIT* — ***September 2021 - Present***

- Analyzed financials of publicly traded companies discovered through industry/market research to find viable, long-term growth investment opportunities to increase the value of the club's stock portfolio. (Bloomberg Terminal, EDGAR)

Other Commitments: Women in Computing Society Member, JerseyCares volunteer, NJIT Kids Who Code tutor,

SKILLS

Technology: Python (PyTorch, Numpy, Scikit-Learn, Pandas), C, C++, C#, Java, OOP, JavaScript, HTML5, CSS, SQL, MySQL, PostgreSQL, PHP, Nodejs, Express, React, RESTful APIs, Jira, Tableau, 3D modeling, creative arts

Certifications: Microsoft Office Specialist Word 2016-2019, Microsoft Office Specialist PowerPoint 2016-2019